

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Hard

Question

The function f is defined by $f(x) = a\sqrt{x+b}$, where a and b are constants. In the xy -plane, the graph of $y = f(x)$ passes through the point $(-24, 0)$, and $f(24) < 0$. Which of the following must be true?

Answer

- A. $f(0) = 24$
- B. $f(0) = -24$
- C. $a > b$
- D. $a < b$

Correct Answer: D

Rationale

Choice D is correct. It's given that $f(24) < 0$. Substituting 24 for $f(x)$ in the equation $f(x) = a\sqrt{x+b}$ yields $f(24) = a\sqrt{24+b}$. Therefore, $a\sqrt{24+b} < 0$. Since $\sqrt{24+b}$ can't be negative, it follows that $a < 0$. It's also given that the graph of $y = f(x)$ passes through the point $(-24, 0)$. It follows that when $x = -24$, $f(x) = 0$. Substituting -24 for x and 0 for $f(x)$ in the equation $f(x) = a\sqrt{x+b}$ yields $0 = a\sqrt{-24+b}$. By the zero product property, either $a = 0$ or $\sqrt{-24+b} = 0$. Since $a < 0$, it follows that $\sqrt{-24+b} = 0$. Squaring both sides of this equation yields $-24+b = 0$. Adding 24 to both sides of this equation yields $b = 24$. Since $a < 0$ and b is 24 , it follows that $a < b$ must be true.

Choice A is incorrect. The value of $f(0)$ is $a\sqrt{b}$, which must be negative.

Choice B is incorrect. The value of $f(0)$ is $a\sqrt{b}$, which could be -24 , but doesn't have to be.

Choice C is incorrect and may result from conceptual or calculation errors.